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第二章、第三章、第四章课后作业

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第二章课后作业：

2-2. 说明MCS-51单片机的引脚$\overline{EA}$的作用，该引脚接高电平和接低电平时各有何功能？

答：①$\overline{EA}$引脚为外部程序存储器地址允许输入/编程电源输入

②第一功能用于选择访问的外部程序存储器，第二功能是当对单片机内部的EPROM编程时，给此引脚施加21V的编程电源。

③$\overline{EA}$接地时，只访问外部程序储存器，$\overline{EA}$接+5V高电平时，先访问内部程序存储器；当地址超过4KB时，将自动转去执行外部程序存储器内的程序。

2-3. MCS-51单片机的引脚中有多少根I/O线？它们与单片机对外的地址总线和数据总线之间有什么关系？其地址总线和数据总线各有多少位？对外可寻址的地址空间有多大？

答：有P0~P3共4根I/O线，在有片外扩展存储器的系统中，P2口作为高8位地址总线，P0口作为低8位地址总线，两者共同组成16位的地址总线，可寻址64KB的地址空间，P0口也作为8位数据总线，实现数据传输。地址总线为16位，数据总线为8位，可寻址64KB地址空间。

2-4. 8051单片机的控制总线信号有哪些？各有何作用？

答：①RST/V_{PD}引脚：第一功能是复位信号输入端，高电平有效，第二功能是备用功能输入端；②$\overline{PSEN}$引脚：作为外部程序存储器允许输出控制端，低电平有效，接单片机外部程序存储器的$\overline{OE}$引脚；③ALE/$\overline{PROG}$引脚：第一功能是地址锁存允许，第二功能是编程脉冲输入；④$\overline{EA}$/V_{PP}引脚：第一功能用于选择访问的外部程序存储器，第二功能是对EPROM编程时，施加21V编程电压。

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2-10. 单片机内部RAM中有哪些单元有位地址？哪些单元没有位地址？

答：内部RAM中位寻址区有位地址，片内RAM地址为20H～2FH；而地址为00H～1FH的通用寄存器区和地址为30H～7FH的用户RAM区没有位地址。

2-12. 当计算机复位后，通用寄存器R0～R7的地址分别是多少？若PSW=18H，则R0～R7的地址又是多少？

答：计算机复位后，R0～R7为默认的第0组地址，R0～R7是00H～07H；PSW=18H时，R0～R7为第三组地址，R0～R7是18H～1FH.

2-13. 程序存储器从0开始的一些单元有什么作用？请写出MCS-51子系列单片机中5个中断源的中断入口地址。

答：从0开始的一些单元被用作特定的入口地址。外部中断0，地址为0003H；定时器0（T_0），地址为000BH；外部中断1，地址为0013H；定时器1（T_1），地址为001BH；串行口，地址为0023H。

第三章课后习题

3-3. 指出下列指令中源操作数和目的操作数的寻址方式.

	源操作数	目的操作数
(1) MOV A, #10H	立即数	寄存器寻址
(2) MOV @R1, A	寄存器寻址	寄存器间接寻址.
(3) MOVC A, @A+DPTR	变址寻址	寄存器寻址.
(4) PUSH ACC	直接寻址.	寄存器间接寻址.
(5) MOV C, 20H	直接寻址.	位寻址.
(6) MOV A, 20H	直接寻址.	寄存器寻址.
(7) MOV R0, P1	直接寻址	寄存器寻址.
(8) JC LOOP	寄存器间接寻址	相对寻址.

3-5. 已知(SP)=26H, (PC)=2345H, (24H)=12H, (25H)=34H, (26H)=56H
问此时执行"RET"指令后, (SP)= 24H , (PC)= 5634H.

3-9 分析下面各段程序中每条指令的运行结果.

(1) MOV A, #45H	将45H存入累加器A中, (A)=45H.
MOV R5, #78H	将78H存入R5中, (R5)=78H.
ADD A, R5	将R5中的78H加入A中, (A)=45H+78H=0BDH
DA A	将A中数转为BCD码. (A)=00011000 1001B
MOV 30H, A.	将A中数移入30H地址中. (30H)=189
(2) MOV SP, #50H	将50H移入堆栈指针中 (SP)=50H
MOV A, #12H	将12H移入累加器A中. (A)=12H.
MOV B, #78H.	将78H移入B寄存器中. (B)=78H.
PUSH ACC	将累加器中的数压入堆栈.

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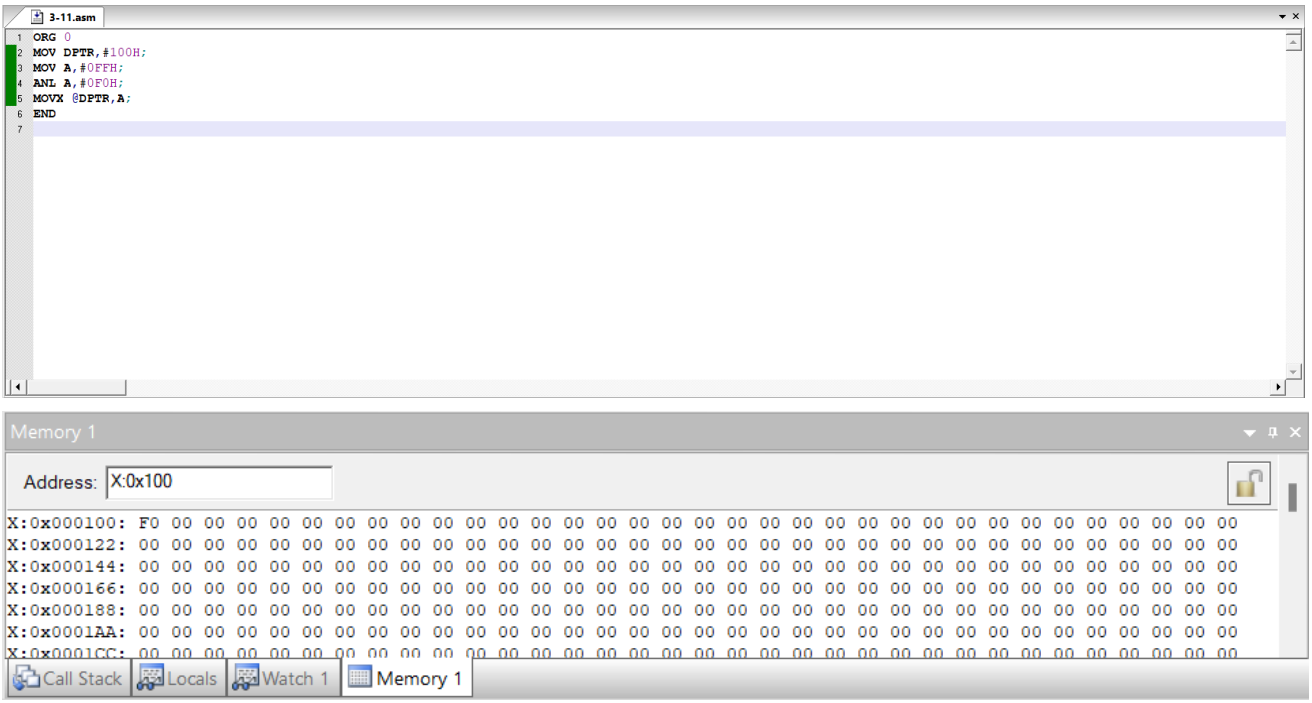
PUSH B	将B寄存器中的数压入堆栈.
POP ACC	将栈顶的数(原B寄存器中的数)弹出至累加器中.
POP B	将原累加器中的数弹出移入B寄存器中.

3-10. 分析下段程序的功能

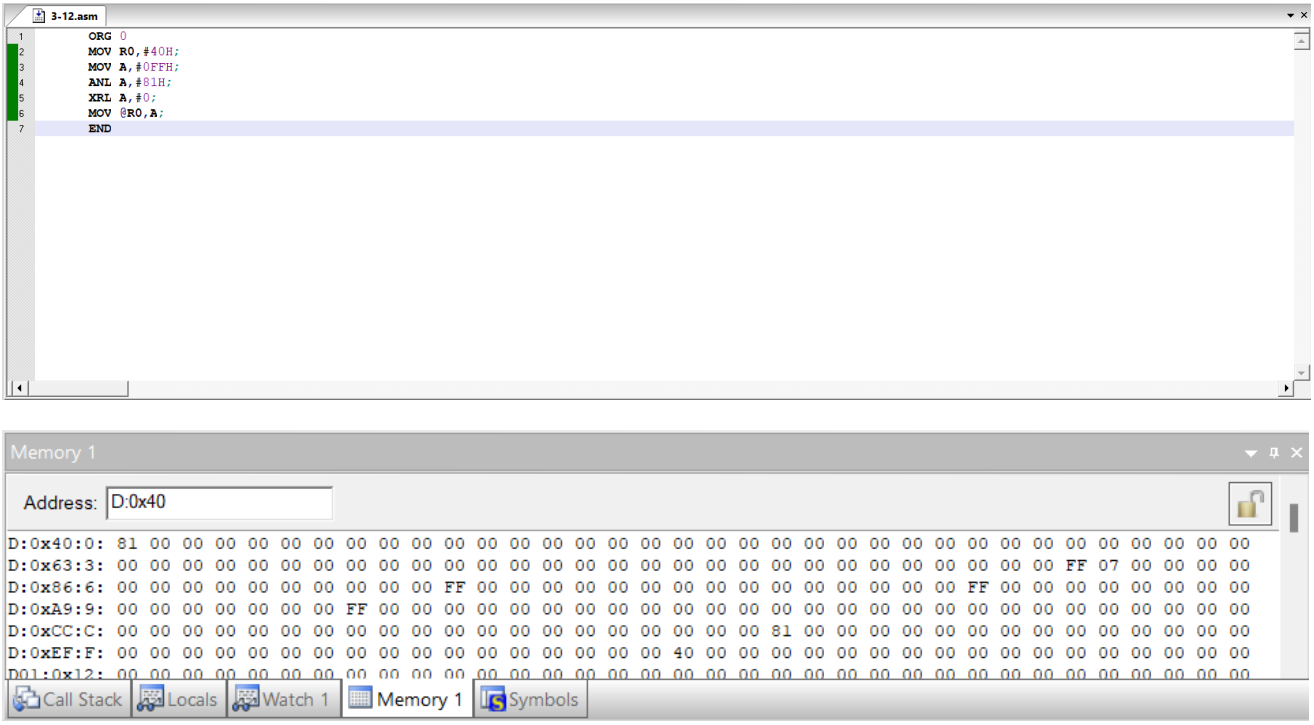
```
        ORG 0          设置程序初始地址为0
START:  MOV R0,#30H    将30H移入R0,作为起始地址.
        MOV R2,#0AH    将0AH移入R2,作为计数器.循环10次.
        DEC R2         R2值减1,循环计数.
        MOV A,@R0      读取R0值并存入累加器A中.
LOOP:   INC R0         R0值增加,访问下一个内存地址.
        MOV 20H,@R0    读取R0的值并存入地址20H中.
        CJNE A,20H,NEXT1  比较A与20H中数的大小,不相等则跳转NEXT1
NEXT1:  JNC NEXT2      若CY为0.即CJNE无进位.则跳转NEXT2.
        MOV A,@R0      将R0中的数存入A中.
NEXT2:  DJNZ R2,LOOP   R2值减小,并在R2不为0时返回循环.
        MOV R7,A       将A中的数移入R7中.
        END            程序结束.
```

分析:这个程序读取了内部RAM中从30H开始的10个单元,并复制其中内容移入20H中,再与原始数据进行比较,若大于则移入R7中反复循环,若小于则重新赋值修正再移入R7,这样可以保证将30H开始的10个地址单元中的最大值均移入R7中。

3-11


3-11.asm
ORG 0
MOV DPTR,#100H;
MOV A,#0FFH;
ANL A,#0F0H;
MOVX @DPTR,A;
END
Memory 1
Address: X:0x100
X:0x000100: F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
X:0x000122: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
X:0x000144: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
X:0x000166: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
X:0x000188: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
X:0x0001AA: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Call Stack
Locals
Watch 1
Memory 1

3-12


3-12.asm
ORG 0
MOV R0,#40H;
MOV A,#0FFH;
ANL A,#81H;
XRL A,#0;
MOV @R0,A;
END
Memory 1
Address: D:0x40
D:0x40:0: 81 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0x63:3: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 07 00 00 00
D:0x86:6: 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00
D:0xA9:9: 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0xCC:C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 81 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0xEF:F: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Call Stack
Locals
Watch 1
Memory 1
Symbols

4-2

```
4-2.asm*
        ORG 0
START:
        MOV DPTR, #1000H
        MOV R0, #30H
        MOV R7, #20
        MOV A,#0


LOOP:
        MOVC A, @A+DPTR
        MOVX @R0, A
        INC DPTR
        INC R0
        CLR A
        DJNZ R7, LOOP

        ORG 1000H
        DB 10,11,15,16
        END
```

4-3

```
4-3.asm
        ORG 0
START:MOV DPTR,#2000H;
      MOV A,#12H;
      MOV R7,#255;
LOOP1:MOVX @DPTR,A;
      INC DPTR;
      DJNZ R7,LOOP1;
      END
```

```
Memory 1
Address: X:0x2000
X:0x002000: 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12
X:0x002022: 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12
X:0x002044: 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12
X:0x002066: 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12
X:0x002088: 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12
X:0x0020AA: 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12
X:0x0020CC: 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12 12
Call Stack | Locals | Watch 1 | Memory 1 | Symbols
```

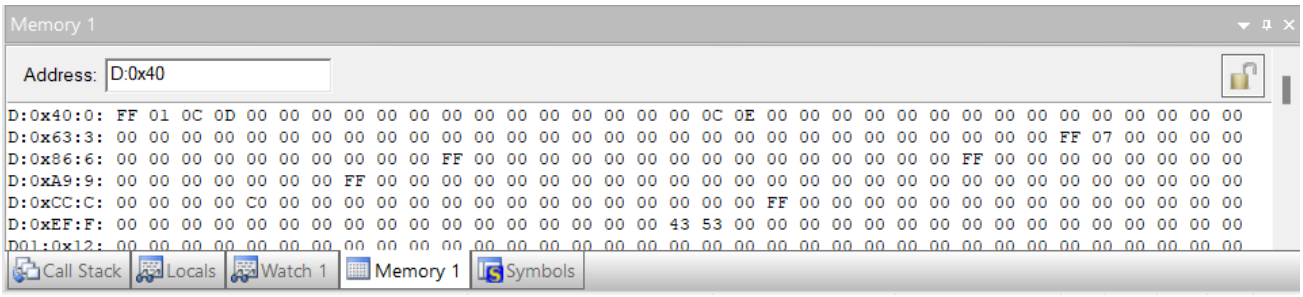
4-5

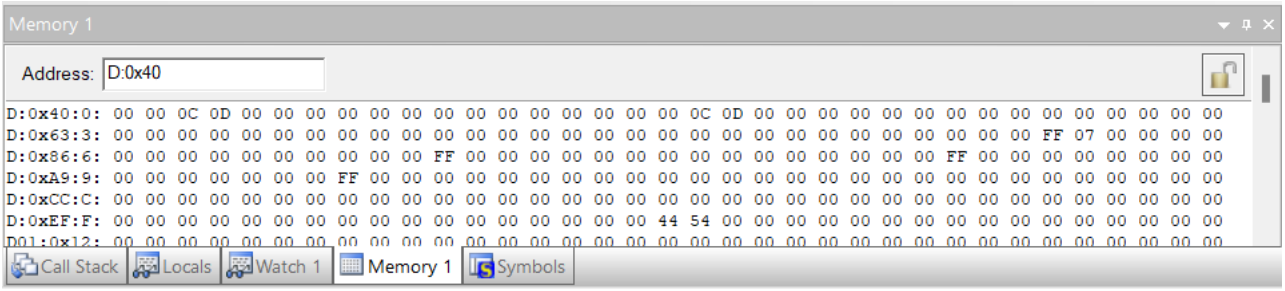
字符串不相等时，输出结果为-1

```
4-5.asm
        ORG 0
START:
        MOV 41H,#2
        MOV 42H,#12
        MOV 43H,#13
        MOV  52H,#12
        MOV 53H,#14
        MOV  R0 ,#42H
        MOV R1,#52H

LOOP:
        MOV A,@R0
        SUBB A,@R1
        JNZ LOOP1
        INC R0
        INC R1
        DJNZ 41H,LOOP
        MOV 40H,#0
        SJMP FINISH
LOOP1:
        MOV 40H,#0FFH
        SJMP FINISH
FINISH:
```

结果如下图所示



字符串相等时，输出结果为 0

结果如下图所示